



Lakshya JEE 2.0 (2024)

Indefinite Integration

DPP-01

1. $\int \frac{1 - \tan^2 x}{1 + \tan^2 x} dx$

2. $\int \frac{dx}{1 + \cos x}$

3. $\int \frac{dx}{\sin^2 x \cos^2 x}$

4. $\int \frac{x^3}{x+1} dx$

5. $\int \frac{x^4 + x^2 + 1}{2(1 + x^2)} dx$

6. $\int \frac{x^4 + 3x^2 + 1}{\sqrt{x}} dx$ equals

(1) $\frac{2}{7}x^{\frac{7}{2}} + 2x^{\frac{3}{2}} + x^{-\frac{1}{2}} + C$

(2) $\frac{2}{9}x^{\frac{7}{2}} + \frac{2}{3}x^{\frac{3}{2}} + \sqrt{x} + C$

(3) $\frac{2}{5}x^{\frac{5}{2}} + \frac{3}{2}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C$

(4) $\frac{2}{9}x^{\frac{9}{2}} + \frac{6}{5}x^{\frac{5}{2}} + 2x^{\frac{1}{2}} + C$

7. $\int \frac{2 + 3x^2}{x^2(1 + x^2)} dx =$

(1) $\tan^{-1}x + \frac{2}{x} + c$

(2) $\frac{2}{x} - \tan^{-1}x + c$

(3) $\tan^{-1}x - \frac{1}{x} + c$

(4) $\tan^{-1}x - \frac{2}{x} + c$

8. $\int \frac{x^4 + x^2 + 1}{x^2 - x + 1} dx$ is equal to

1) $\frac{x^3}{3} - \frac{x^2}{2} + x + C$

(2) $\frac{x^3}{3} + \frac{x^2}{2} + x + C$

(3) $\frac{x^3}{3} - \frac{x^2}{2} - x + C$

(4) $\frac{x^3}{3} + \frac{x^2}{2} - x + C$

9. $\int 2^x \cdot e^x dx =$

(1) $\frac{(2e)^x}{\ln(2e)} + c$ (2) $\frac{e^2}{\ln(2e)} + c$

(3) $\frac{(2e)^x}{\ln 2x} + c$ (4) None of these

10. $\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx = \frac{2 \cdot a^x}{\ln a} - \frac{1}{5} \frac{b^x}{\ln b} + c$, where c is an integration constant then $10ab =$

(1) -1 (2) 0

(3) 1 (4) 2

11. $\int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx =$

(1) $\tan x + \tan^{-1}x + c$

(2) $\tan^{-1}x - \tan x + c$

(3) $\tan x - \tan^{-1}x + c$

(4) $c - \tan x - \tan^{-1}x$

12. $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2\sin^2 2x} dx =$

(1) $2\cos x + c$

(2) $-2\sin x + c$

(3) $2\sin x + c$

(4) $-2\cos x + c$



13. $\int \frac{x^6 - 1}{x^2 + 1} dx =$

(1) $-\frac{x^5}{5} - \frac{x^3}{3} + x - 2 \tan^{-1} x + c$

(2) $\frac{x^5}{5} + \frac{x^3}{3} + x - 2 \tan^{-1} x + c$

(3) $\frac{x^5}{5} - \frac{x^3}{3} + x - 2 \tan^{-1} x + c$

(4) $\frac{x^5}{5} - \frac{x^3}{3} + x + 2 \tan^{-1} x + c$

14. $\int \frac{e^{2x} - 1}{e^x} dx$

15. $\int \frac{(\sqrt{x} + 1)(x^2 - \sqrt{x})}{x\sqrt{x} + x + \sqrt{x}} dx$

16. $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx$

17. A function g defined for all positive real numbers, satisfies $g'(x^2) = x^3$ for all $x > 0$ and $g(1) = 1$. Compute $g(4)$.

18. $\int \sin x \cos x \cos 2x \cos 4x dx$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

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|---|---------------------------------|
| 1. $(\frac{1}{2} \sin 2x + c)$ | 10. (3) |
| 2. $(\tan(x/2) + C)$ | 11. (3) |
| 3. $-2 \cot 2x + c$ | 12. (4) |
| 4. $(\frac{x^3}{3} - \frac{x^2}{2} + x - \ln(x+1) + C)$ | 13. (3) |
| 5. $\frac{1}{2} \left[\frac{x^3}{3} + \tan^{-1} x \right] + C$ | 14. $e^x + e^{-x} + C$ |
| 6. (4) | 15. $\frac{x^2}{2} - x + C$ |
| 7. (4) | 16. $-\frac{\cos 4x}{8} + C$ |
| 8. (2) | 17. $\frac{67}{5}$ |
| 9. (1) | 18. $-\frac{1}{64} \cos 8x + C$ |



PW Web/App - <https://smart.link/7wwwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>